ProgV-Trace: An LLM-Driven Verification Framework for Evidence-Grounded Video Misinformation Attribution

Yifan Xiao
Peking University
Beijing, China
2401110754@stu.pku.edu.cn

Shijie Li
China Southern Power Grid Company Limited
Guangzhou, China
geyh@csg.cn

Yu Huang*
Peking University
Beijing, China
hy@pku.edu.cn

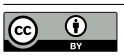
Abstract

Video-based misinformation—assembled from authentic footage, fabricated captions, and manipulated speech—spreads rapidly across platforms, yet existing multimodal detectors typically produce only a binary verdict or a post-hoc rationale without determining *which* specific element across video, speech, and text is responsible for making the narrative false, or *to what degree*. Current approaches treat misinformation analysis as a single-step classification augmented with explanations, collapsing the structured, multi-step reasoning that human fact-checkers perform into a single inference pass.

We formalize *LLM-Driven Evidence-Grounded Multimedia Misinformation Attribution* (L-EMMA), a task that jointly requires veracity classification, fine-grained falsity grounding, evidence-backed provenance chains, and quantified interventional responsibility scores. To address L-EMMA, we propose ProgV-Trace, a framework in which a large language model generates an executable *verification program*—a typed operator sequence in a domain-specific language—that decomposes claims, orchestrates evidence retrieval, dynamically expands a temporal-multimodal provenance graph, and plans minimal counterfactual interventions to isolate each element's contribution to the misinformation verdict. A program-to-student distillation stage transfers the LLM's structured reasoning into a compact verifier for deployment without LLM inference. We further introduce a *Program-Hard Challenge Protocol* with fresh, semantically coordinated, and multi-step-verification samples that stress-test generalization beyond standard benchmarks.

Extensive experiments on GroundLie360, FakeVV, and FakeSV demonstrate that ProgV-Trace achieves 87.2% F1 and 71.4% Resp-NDCG, surpassing the strongest baseline by +4.5 F1 and +24.2 Resp-NDCG points—of which a same-model ablation confirms +21.8 points stem from the interventional attribution method itself. The distilled student retains over 96% attribution quality at 196 ms per sample, and responsibility scores agree with human judgments 84.9% on average across five manipulation types.

*Corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License.
MM '26, Rio de Janeiro, Brazil
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2213-4/2026/11
https://doi.org/10.1145/3767308.3836284

CCS Concepts

• **Information systems** → **Multimedia and multimodal retrieval**; • **Computing methodologies** → *Natural language processing*; *Neural networks*.

Keywords

video misinformation, multimedia reasoning, provenance graph, counterfactual reasoning, multimodal attribution, evidence-grounded detection

ACM Reference Format:
Yifan Xiao, Shijie Li, and Yu Huang. 2026. ProgV-Trace: An LLM-Driven Verification Framework for Evidence-Grounded Video Misinformation Attribution. In *Proceedings of the 34th ACM International Conference on Multimedia (MM '26), November 10–14, 2026, Rio de Janeiro, Brazil.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3767308.3836284

1 Introduction

Video-based misinformation—where misleading narratives are assembled from authentic footage, fabricated captions, manipulated speech transcripts, and deceptive on-screen overlays—has emerged as one of the most pressing challenges in multimedia content understanding [49, 61, 71]. Major platforms process vast volumes of video uploads daily, yet verifying a single video claim requires cross-referencing multiple modalities against external sources, a process that demands significantly more analyst effort than text-only fact-checking [59, 71]. Multimodal misinformation spreads substantially faster than text-only counterparts, partly because viewers perceive video as inherently more authentic [13, 19, 42]. While substantial progress has been made through external evidence retrieval [6, 38], agentic tool invocation [9, 29], and manipulation grounding [64, 69, 70], existing approaches share a fundamental limitation: they treat misinformation analysis as a single-step classification problem—sometimes augmented with post-hoc explanations—rather than as a structured, multi-step verification process that mirrors how human fact-checkers actually work.

Consider how a human fact-checker approaches a suspicious video: they first decompose the narrative into verifiable sub-claims—who, what, when, where—then systematically search for corroborating or contradicting evidence, expand their investigation when initial results are ambiguous, and finally determine which specific fabrication is most responsible for making the story misleading. This process is inherently *programmatic*: it involves conditional branching, iterative evidence gathering, and explicit stopping criteria. Yet current systems, even those employing chain-of-thought reasoning [62, 67] or agentic tool use [9, 29], collapse this structured process into a single inference pass that outputs a verdict and

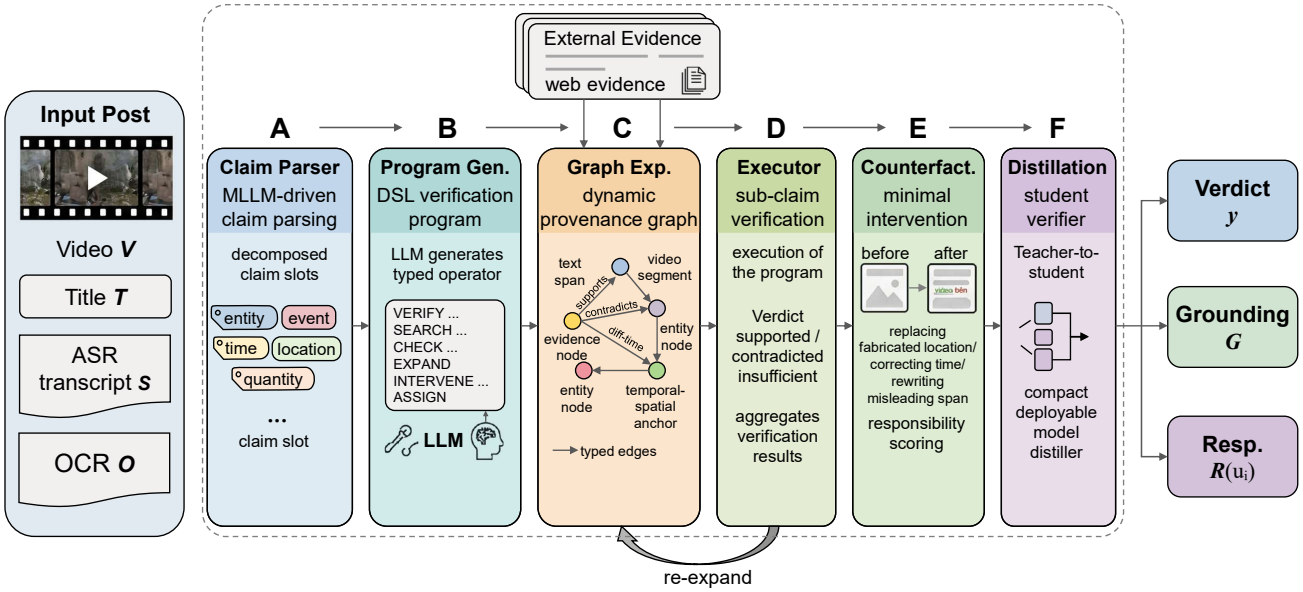


Figure 1: Overview of ProgV-Trace. An LLM-driven verification framework that parses multimodal claims into structured slots (Module A), generates an executable verification program (Module B), uses it to orchestrate evidence retrieval and dynamic provenance graph expansion (Module C), executes sub-claim verification with verdict aggregation (Module D), performs LLM-planned minimal counterfactual interventions for responsibility attribution (Module E), and distills the full reasoning trace into a compact student verifier (Module F).

optional explanation without making the verification logic explicit, executable, or reusable.

We argue that the LLM should be elevated from a passive classifier to an active *verification program generator*: instead of directly predicting verdicts, the model should produce a structured, executable plan specifying what to verify, how to gather evidence, when to expand, and where to intervene.[1] This shift addresses three gaps: the absence of executable verification representations beyond free-text rationales, the lack of program-guided dynamic evidence organization that adapts to each claim's complexity, and the need for LLM-planned interventions that produce genuine interventional responsibility scores rather than correlational attention weights [33, 40, 46].

To realize this vision, we propose ProgV-Trace, an LLM-driven verification framework for evidence-grounded video misinformation attribution. Given a multimodal post, ProgV-Trace first parses claims into structured slots via MLLM-driven extraction, then generates a *verification program*—a sequence of typed operators in a domain-specific language that specifies sub-claim decomposition, evidence query templates, graph expansion instructions, contradiction checks, and intervention plans. This program drives dynamic expansion of a temporal-multimodal provenance graph, connecting claim elements to external evidence through typed relations. A program executor verifies each sub-claim and aggregates verdicts with uncertainty tracking, while an LLM-guided counterfactual module performs minimal interventions to compute interventional responsibility scores. A program-to-student distillation stage transfers reasoning traces into a compact verifier for efficient deployment. We further introduce a *Program-Hard Challenge Protocol* with fresh, semantically coordinated, and multi-step-verification samples that stress-test generalization beyond standard benchmarks.

Our contributions are summarized as follows:

- **(C1)** We formalize L-EMMA and propose ProgV-Trace, where an LLM generates executable verification programs for evidence retrieval, graph expansion, and interventional attribution (Section 4).
- **(C2)** We introduce program-guided provenance graph expansion and LLM-planned counterfactual intervention with program-to-student distillation (Sections 4.3–4.6).
- **(C3)** We propose a Program-Hard Challenge Protocol and demonstrate on three benchmarks that ProgV-Trace outperforms state-of-the-art baselines across detection, grounding, and attribution (Section 5).

2 Related Work

2.1 MLLM-Based Fact-Checking with External Evidence

Recent work has increasingly leveraged multimodal large language models with external knowledge sources to verify claims, building on progress in text-based fact verification [13, 53],

[1] We use "interventional" to denote model-internal perturbation-based attribution, distinct from formal causal identification in Pearl's framework [41].

retrieval-augmented reasoning [24], and multimodal understanding [3, 27, 43]. DEFAME [6] introduces a modular six-stage pipeline in which an MLLM decomposes claims, retrieves evidence from search engines and fact-checking databases, and synthesizes a verdict through multimodal experts, achieving strong zero-shot performance across multiple benchmarks. CMIE [25] combines MLLM insights with selectively fused external evidence, producing explainable verdicts for out-of-context misinformation. Aletheia [38] emphasizes evidence quality over quantity through a retrieval-then-verify paradigm that filters noisy web results before reasoning. While these methods demonstrate the value of MLLM-driven evidence reasoning, they treat evidence as flat supporting material for a single classification step rather than as structured inputs to a multi-step verification program that can dynamically adapt its evidence strategy based on intermediate findings.

2.2 Agentic and Tool-Augmented Misinformation Verification

A growing line of work frames misinformation detection as an agentic task where models invoke external tools to gather and process evidence. T2Agent [9] combines an extensible toolkit with Monte Carlo tree search to dynamically collect evidence from multiple sources, serving as a training-free detector for mixed-source misinformation. FactGuard [29] extends this paradigm to video misinformation by integrating external tool invocation—search engines, reverse image lookup, temporal analyzers—with reinforcement learning to train step-by-step verification chains, representing one of the strongest agentic baselines for video fact-checking. Visual programming [14, 51] does emit executable programs, but for compositional perception rather than evidence-grounded verification; Tree of Thoughts [65] searches reasoning paths without emitting a program at all. In agentic verification the logic thus stays implicit in the chain-of-thought or action sequence, not a typed program with conditional branching and stopping criteria. As a result, it is difficult to audit which verification steps were executed, understand why the model stopped searching, or transfer the verification strategy to new claims without re-running the full inference pipeline.

2.3 Video Misinformation Detection and Grounding

Video misinformation presents unique challenges due to the temporal dimension and the interplay between visual content, speech, and overlaid text [1, 19, 21, 23, 72]. GroundLie360 [64] formally defines Grounding Multimodal Misinformation across title, speech, and visual tracks, providing 10K annotated video-text pairs that advance beyond image-only benchmarks [2, 5, 34, 36]. Fact-R1 [67] proposes a reinforcement learning framework that produces step-by-step reasoning chains for video misinformation detection, accompanied by the large-scale FakeVV dataset. FakeSV-VLM [57] integrates vision-language models for short social media video detection, leveraging advances in vision-language pretraining [27, 43]. These methods advance detection accuracy and grounding precision, but their outputs remain at the level of verdicts and localized spans without quantifying how much each grounded element contributes to making the narrative misleading—the differential responsibility information that content moderators need to prioritize their response.

2.4 Dual-Source and Forensic Reasoning

D-SECURE [4] combines internal manipulation cues with external evidence reasoning in a dual-source architecture, demonstrating that neither source alone is sufficient for robust detection. REFORM [69] shifts manipulation detection from outcome-based classification to process-based forensic reasoning through a three-stage curriculum of rationale induction, reasoning alignment, and consistency refinement, achieving strong generalization across DGM4 and MMFakeBench. ASAP [70] and CSCL [28] advance alignment-based prompting and consistency-supervised contrastive learning for fine-grained manipulation grounding, respectively. These hybrid and forensic methods demonstrate the value of structured multi-source integration [8, 60, 66] and process-level supervision, yet they produce a single enhanced verdict without making the reasoning process executable, auditable, or transferable to a lightweight student model. Our work extends this philosophy by generating an explicit verification program that serves as both the reasoning engine and the supervision signal for distillation.

2.5 Challenge Benchmarks for Realistic Misinformation

Recent benchmarks target realistic, mixed-source misinformation: MMFakeBench [31] evaluates LLMs on diverse forgery types, SAMM [48] focuses on semantically aligned manipulation, and OmniFake [26] pushes toward multi-generator realism. MUSE [40] demonstrates that many detectors exploit surface-level shortcuts, underscoring the need for intervention-based evaluation. Post-hoc attribution methods [33, 44, 46] measure correlational importance without leveraging structured evidential relationships. Counterfactual reasoning has been applied in VQA debiasing [37], scene graph generation [52], and visual commonsense [58], while CIMDD [30] and TGFC-SVFN [74] apply causal intervention to misinformation detection. Neuro-symbolic program synthesis [7, 20, 35, 50] inspires our adaptation of executable programs to misinformation attribution. These benchmarks motivate our Program-Hard Challenge Protocol.

Existing work covers retrieval-enhanced verification, agentic tool use, video-level grounding, and reasoning-driven forensics, but no prior method unifies executable verification programs, program-guided provenance graph expansion, minimal counterfactual interventions for interventional attribution, and program-to-student distillation.

3 Task Formulation

We formalize *LLM-Driven Evidence-Grounded Multimedia Misinformation Attribution* (L-EMMA) as a task that extends conventional misinformation detection with executable verification, fine-grained grounding, and quantified responsibility scoring. To motivate the task, consider a viral social media post where authentic flood footage is paired with a fabricated caption claiming the disaster struck a different city last week. A human fact-checker would not simply label this as "fake"—they would decompose the claim into verifiable sub-elements, search for the original source, check

temporal and geographic consistency, and determine that the location fabrication is the primary driver. This structured, multi-step process is what L-EMMA formalizes.

Given a social media post containing a video V, a speech transcript S from ASR, a user-authored title $\mathcal{T}$, optional on-screen text O from OCR, and retrieved external evidence $E = \{e_1, \ldots, e_K\}$, the system must jointly produce six outputs: a **veracity label** y indicating the misinformation type; a **verification program** P_{ver} specifying the executable verification plan in a domain-specific language; **textual grounding** G_t marking token-level spans in $\mathcal{T}$, S, and O that contribute to the false narrative; **temporal grounding** G_v^{temp} identifying misleading video segments; an **evidence chain** C recording which claim element is linked to which evidence through what relation type; and a **responsibility score map** $R : \mathcal{U} \to [0, 1]$ assigning each candidate element an interventional responsibility score. Returning to our running example, the system should generate a verification program that checks the claimed location against geolocated evidence, identifies the temporal mismatch, and determines that the geographic fabrication bears responsibility of 0.72 while the temporal mismatch contributes 0.58.

The L-EMMA task differs from prior formulations in three key respects. First, the inclusion of the verification program P_{ver} as an explicit output makes the reasoning process auditable and transferable—unlike chain-of-thought rationales that are generated post-hoc, P_{ver} specifies typed operators that drive the actual verification execution. Second, the responsibility score R is defined interventionally: it reflects the change in system behavior when an element is systematically perturbed under LLM-planned minimal interventions, rather than attention weights [55] or gradient norms [46]. Third, unlike detection methods [29, 67] that produce only a verdict, or grounding tasks [47, 64] that additionally localize suspicious regions, L-EMMA requires all six outputs simultaneously—ensuring that every verdict is accompanied by an executable program, traceable evidence, and quantified attribution.

4 Method

Figure 1 illustrates the overall architecture of ProgV-Trace, which processes a multimodal post through six stages: parsing claims into structured slots (Module A), generating an executable verification program (Module B), expanding a provenance graph guided by the program (Module C), executing sub-claim verification with verdict aggregation (Module D), performing LLM-planned minimal counterfactual interventions for responsibility attribution (Module E), and distilling the full reasoning trace into a compact student verifier (Module F).

4.1 Multimodal Claim Parser

The first module transforms raw multimodal inputs into structured claim slots—the atomic units that the verification program will operate on. The key idea is to decompose a complex multimodal narrative into individually verifiable factual elements, mirroring how a human fact-checker would break apart a claim before investigating each piece.

For the textual modalities—title $\mathcal{T}$, ASR transcript S, and OCR text O—we employ a BERT-based encoder [10] with a sequence labeling head to extract six slot types: entities, events, temporal markers, locations, quantities, and causal relations. For the visual modality, we uniformly sample N_f keyframes from video V and encode each with CLIP ViT-L/14 [11, 43] to produce frame-level features $\{\mathbf{f}_1, \ldots, \mathbf{f}_{N_f}\}$. Temporal segmentation partitions V into semantically coherent segments $\{v_1^{\text{seg}}, \ldots, v_L^{\text{seg}}\}$, each aligned with corresponding ASR spans. The output is a unified set of claim elements $\mathcal{U} = \{u_1, \ldots, u_N\}$, where each u_i carries its slot type, source modality, position span, and a dense feature representation $\mathbf{h}_i \in \mathbb{R}^{768}$. When ASR produces no transcript—as in silent or music-only videos—the model operates on title, OCR, and video modalities alone.

4.2 Verification Program Generator

This is the core LLM-driven module that distinguishes ProgV-Trace from prior detection and agentic approaches. Instead of letting the model directly predict a verdict, we ask the LLM to generate an explicit *verification program*—a structured sequence of typed operators in a domain-specific language (DSL) that specifies exactly what to verify, how to gather evidence, when to expand the investigation, and where to intervene.

The DSL defines six operator types: `VERIFY` checks a specific claim attribute against available evidence; `SEARCH` triggers retrieval of a specified evidence type; `EXPAND` adds nodes and edges to the provenance graph around a specified focus point; `CHECK` evaluates consistency between two elements along a specified dimension; `INTERVENE` specifies a minimal counterfactual perturbation; and `ASSIGN` computes responsibility for a target element. Given the parsed claim elements $\mathcal{U}$ from Module A, the LLM generates a program $P_{\text{ver}} = (o_1, o_2, \ldots, o_M)$ where each o_j is a typed operator with structured arguments. For example, a program for the flood video scenario might contain:

```
VERIFY(entity="flood", time="last week", loc="CityX")
SEARCH(type=reverse_image, query=keyframe_3)
CHECK(temporal, claimed="last week", evidence="Mar 2024")
EXPAND(node=evidence_2, relation=same_event)
INTERVENE(span=title_loc, action=replace, target="CityY")
ASSIGN(element=title_loc)
```

Each operator has a formally defined type signature specifying its inputs, preconditions, and outputs:

```
VERIFY(attr:SlotType, value:str) -> Bool|Unknown
SEARCH(type:EvidenceType, query:str) -> List[Node]
EXPAND(node:NodeID, relation:EdgeType) -> Graph
CHECK(dim:Dimension, claimed:str, evidence:str) -> Match
INTERVENE(span:SpanID, action:Action, target:str) -> Delta
ASSIGN(element:ElementID) -> Score[0,1]
```

where `SlotType` $\in$ {entity, event, time, location, quantity, relation}, `EvidenceType` $\in$ {news, fact_check, reverse_image, knowledge_base}, `EdgeType` matches the seven provenance relations, and `Action` $\in$ {replace, remove, rewrite}. The complete BNF grammar, operator semantics, and well-formedness constraints are released with our code. The executor processes operators sequentially, halting when a `STOP` condition is met—either all sub-claims are verified, the evidence budget (K=10 retrievals) is exhausted, or the graph reaches 50 nodes. Invalid operator sequences—such as `ASSIGN` before any `SEARCH`—trigger a fallback to direct CoT reasoning (observed in 7.7% of samples). The program generation uses structured output prompting with few-shot DSL examples, ensuring deterministic downstream execution. Figure 2 illustrates the flow.

Table 1: End-to-end comparison on GroundLie360. Best in bold, second-best underlined. †: post-hoc attention attribution. ProgV-Trace: mean$_{\pm\text{std}}$ over 3 seeds; all improvements over D-SECURE significant at $p<0.01$ (paired bootstrap). Prog-SR: program execution success rate (%).

Method	ACC	F1	Token-F1	Temp-IoU	Resp-NDCG	Prog-SR
LLM-driven / agentic verification						
DEFAME† [6]	80.4	78.9	58.7	—	44.1	—
CMIE† [25]	79.7	77.6	60.2	—	42.3	—
T2Agent† [9]	82.1	80.3	—	—	45.6	—
FactGuard† [29]	83.2	81.5	—	—	43.8	—
Video misinformation detection						
FakeMark [64]	78.3	76.1	62.4	51.7	—	—
Fact-R1† [67]	81.6	79.8	—	48.3	41.2	—
FakeSV-VLM† [57]	76.8	74.2	—	44.1	36.5	—
Reasoning / forensic						
D-SECURE† [4]	84.1	82.7	65.3	55.8	47.2	—
REFORM† [69]	76.4	74.3	67.5	56.8	—	—
ProgV-Trace (ours)	**88.7**$_{\pm0.4}$	**87.2**$_{\pm0.5}$	**72.8**$_{\pm0.8}$	**63.1**$_{\pm1.1}$	**71.4**$_{\pm0.9}$	**92.3**

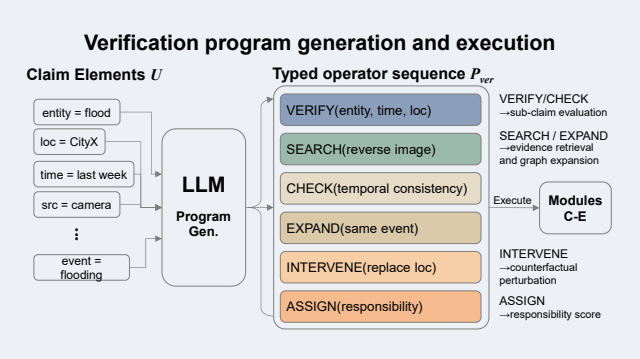


Figure 2: Verification program generation and execution. The LLM generates a typed operator sequence P_{ver} from parsed claim elements. Each operator has well-defined semantics: `VERIFY/CHECK` trigger sub-claim evaluation, `SEARCH/EXPAND` drive evidence retrieval and graph expansion, `INTERVENE` plans counterfactual perturbations, and `ASSIGN` computes responsibility scores.

4.3 Program-Guided Evidence Retrieval and Graph Expansion

The verification program drives dynamic construction and expansion of a temporal-multimodal provenance graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Unlike static graph construction approaches [12] that build the complete graph before reasoning, our graph grows incrementally as the program executes `SEARCH` and `EXPAND` operators.

The graph contains five types of nodes: textual span nodes from title, speech, and OCR; video segment nodes representing temporal portions of the visual content; external evidence nodes from retrieved documents; entity nodes for named entities shared across modalities; and temporal-spatial anchor nodes for dates, locations, and events. The edges encode seven relation types: `supports` and `contradicts` connect evidence to claims based on semantic entailment; `same-event` links elements referring to the same underlying event; `different-time` and `different-location` mark temporal or spatial discrepancies; `recontextualizes` indicates authentic content used in a misleading new context; and `source-of` traces visual content provenance to its original publication. Figure 4 illustrates the graph schema.

When the program executes a `SEARCH` operator, the system retrieves evidence documents from fact-checking databases and web search. Retrieved documents are added as evidence nodes with edges to the relevant claim elements. When an `EXPAND` operator fires, relation-specific binary classifiers (two-layer MLPs trained on 7.2K annotated pairs from GroundLie360) predict new edges around the specified focus node. The LLM controller decides when to stop expansion based on the program's stop conditions, typically after evidence coverage exceeds a threshold or a maximum of 50 nodes and 200 edges is reached.

A 3-layer Relational Graph Convolutional Network (R-GCN) [45], building on graph neural network foundations [15, 22, 63], propagates information across the provenance graph:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\sum_{r\in\mathcal{R}}\sum_{j\in\mathcal{N}_i^r}\frac{1}{|\mathcal{N}_i^r|}\mathbf{W}_r^{(l)}\mathbf{h}_j^{(l)} + \mathbf{W}_0^{(l)}\mathbf{h}_i^{(l)}\right), \tag{1}$$

where $\mathcal{N}_i^r$ denotes the neighbors of node i under relation r, and $\mathbf{W}_r^{(l)}$ is the relation-specific weight matrix. Features are projected from 768 to 256 dimensions before R-GCN input. The resulting graphs average 22.3 ± 8.1 nodes and 58.2 ± 21.4 edges.

4.4 Program Executor and Verdict Aggregator

The program executor processes each `VERIFY` and `CHECK` operator in P_{ver} to produce sub-claim verdicts. For each sub-claim, the executor evaluates the relevant graph neighborhood: attention-weighted pooling [56] over connected evidence nodes produces a local representation, which a classification head maps to a sub-claim status—supported, contradicted, or insufficient evidence. The LLM

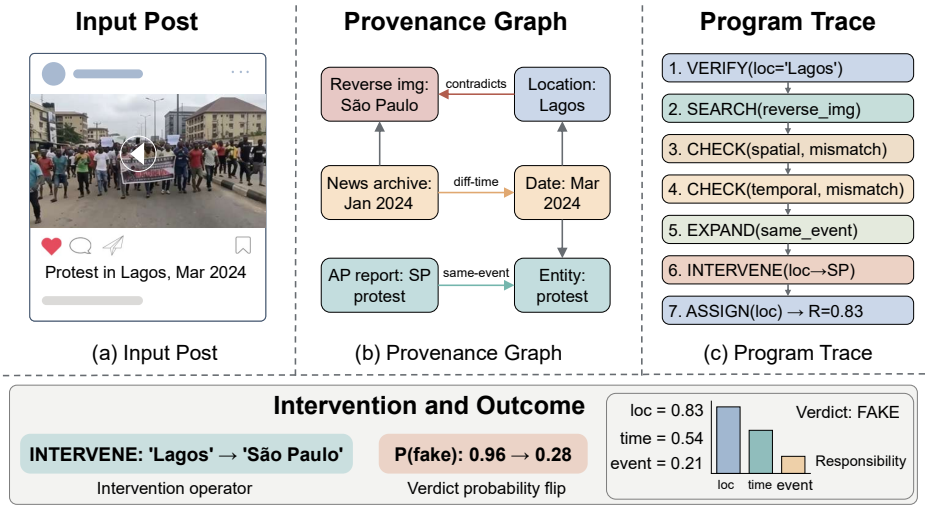


Figure 3: Qualitative case study: a recontextualized protest video. (a) Input post with fabricated caption. (b) Provenance graph connecting claim slots to retrieved evidence via typed edges (contradicts, diff-time, same-event). (c) 7-step verification program trace. (d) The `INTERVENE` operator replaces "Lagos" with evidence-supported "São Paulo," flipping verdict probability from 0.96 to 0.28 and assigning primary responsibility R=0.83 to the location fabrication.

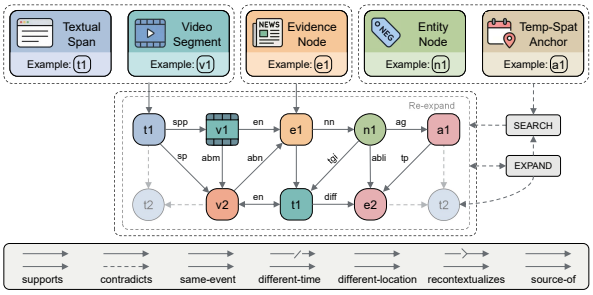


Figure 4: Schema of the temporal-multimodal provenance graph. Five node types (top) are connected via seven typed edges (bottom). The graph is dynamically expanded by the verification program's `SEARCH` and `EXPAND` operators. Avg: 22.3±8.1 **nodes,** 58.2±21.4 **edges.**

controller aggregates sub-claim verdicts into a final verdict y with an uncertainty score, tracking which sub-claims contributed most to the decision and resolving conflicts when evidence is ambiguous. Node-level classifiers produce textual grounding G_t and temporal grounding G_v^{temp}, while the evidence chain C is extracted by tracing the highest-weight paths from grounded claim elements to their connected evidence.

4.5 LLM-Guided Minimal Counterfactual Intervention

The counterfactual module answers L-EMMA's central question: *how much does each content element contribute to the misinformation verdict?* Instead of random masking or uniform perturbation, the LLM generates *minimal fact-flipping interventions* via the program's `INTERVENE` operators—replacing a fabricated location with the evidence-supported location, correcting a temporal mismatch to the verified date, or rewriting an ASR clause to be evidence-consistent.

For each grounding candidate u_i with score above threshold τ_g=0.3, the system applies the LLM-planned intervention, re-runs the R-GCN encoder on the perturbed graph, and computes the responsibility score:

$$R(u_i) = \alpha \cdot \Delta_{\text{verdict}} + \beta \cdot \Delta_{\text{path}} + \gamma \cdot \Delta_{\text{ground}} + \delta \cdot \Delta_{\text{conf}}, \quad (2)$$

where Δ_{verdict} measures the change in verdict probability, Δ_{path} quantifies provenance chain disruption as the Jaccard distance between original and perturbed edge sets, Δ_{ground} captures grounding shift, and Δ_{conf} reflects confidence change. The weights α=0.4, β=0.3, γ=0.2, δ=0.1 reflect decreasing signal directness. Locality of graph message passing restricts re-computation to L-hop neighborhoods, keeping the per-intervention cost proportional to the local subgraph size rather than the full graph.

4.6 Program-to-Student Distillation

To address deployment efficiency and reproducibility concerns, we distill the LLM teacher's structured outputs into a compact student verifier following the knowledge distillation paradigm [16]. The teacher produces five supervision signals: the verification program P_{ver}, evidence chain C, sub-claim verdicts, intervention plans, and responsibility scores. The student—a BERT-base encoder with task-specific heads—learns to predict verdicts, grounding, provenance paths, and intervention sensitivity without requiring LLM inference at test time.

4.7 Training Objectives

ProgV-Trace is trained with five complementary losses: classification loss $\mathcal{L}_{\text{cls}}$ (cross-entropy for veracity), grounding loss $\mathcal{L}_{\text{grd}}$ (binary cross-entropy over token and segment predictions), provenance edge loss $\mathcal{L}_{\text{prov}}$ (relation type supervision), counterfactual consistency loss $\mathcal{L}_{\text{cf}}$ (margin-based pairwise ranking ensuring responsibility ordering matches ground truth with margin δ=0.1), and distillation loss $\mathcal{L}_{\text{distill}}$ (KL divergence between teacher and student logits). The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_1 \mathcal{L}_{\text{grd}} + \lambda_2 \mathcal{L}_{\text{prov}} + \lambda_3 \mathcal{L}_{\text{cf}} + \lambda_4 \mathcal{L}_{\text{distill}}, \tag{3}$$

with λ_1=1.0, λ_2=0.5, λ_3=0.3, λ_4=0.5.

5 Experiments

We evaluate ProgV-Trace on three video misinformation benchmarks and a curated challenge protocol, comparing against 9 strong baselines from 2024–2026.

Datasets. Our primary experiments use GroundLie360 [64], which provides 10K video-text pairs with multi-granularity grounding annotations across title, speech, and visual tracks; FakeVV [68], a large-scale video misinformation dataset with 20K samples and reasoning chain annotations; and FakeSV [42], containing 5K short social media videos with multimodal annotations [39]. All three datasets are publicly available; our experiments use their official train/val/test splits. For external evidence, we retrieve up to K=10 documents per claim from fact-checking databases and web search, following DEFAME's [6] retrieval protocol. On GroundLie360, the retrieval pipeline achieves 78.4% precision@10.

Baselines. We compare against methods from three categories, all published in 2024 or later: *LLM-driven / agentic verification* including DEFAME [6], CMIE [25], T2Agent [9], and FactGuard [29]; *video misinformation detection* including FakeMark [64], Fact-R1 [67], and FakeSV-VLM [57]; and *reasoning / forensic methods* including D-SECURE [4] and REFORM [69]. For baselines without native attribution, we augment them with attention-based attribution post-hoc [33, 46].

Metrics. We report ACC and macro F1 for veracity, Token-F1 for textual grounding, Temporal-IoU for video grounding, Resp-NDCG for responsibility attribution quality [18], and Prog-SR (program execution success rate) measuring the fraction of verification programs that execute to completion without errors. Ground-truth responsibility rankings on GroundLie360 are derived from annotator-labeled misleading spans ordered by manipulation severity (3 annotators, Kendall's τ=0.81). For FakeVV and FakeSV, we construct proxy rankings from dataset-provided grounding labels with ties broken by annotator confidence scores, ensuring independence from our model. We view Resp-NDCG as a necessary but coarse evaluation of attribution quality; for all baselines, attribution scores are computed via the same post-hoc protocol [46] applied uniformly to ensure comparability.

Implementation. The LLM controller uses GPT-4o for program generation; the student verifier uses BERT-base (d=768) with CLIP ViT-L/14 for visual encoding, projected to 256 dimensions. The 3-layer R-GCN uses 256-dim hidden states. Training: 30 epochs, AdamW [32], lr 2×10^{-5}, batch 16, 4×A100 GPUs. Hyperparameters: τ_g=0.3, α=0.4, β=0.3, γ=0.2, δ_{margin}=0.1. Results averaged over 3 seeds; program generation uses greedy decoding (temperature 0) for deterministic output. Hyperparameter sensitivity analysis shows Resp-NDCG varies less than 3 points across tested ranges. For baselines without code, we re-implement from reported architectures.

Table 2: Cross-dataset evaluation on FakeVV and FakeSV. All ProgV-Trace improvements over D-SECURE significant at p<0.05 (paired bootstrap). Resp-NDCG on FakeSV uses proxy labels; "—": no native attribution.

	FakeVV			FakeSV		
Method	ACC	F1	NDCG	ACC	F1	NDCG
Fact-R1	80.2	78.4	40.8	76.1	74.0	37.6
FactGuard	81.9	80.1	41.3	78.4	76.8	38.2
T2Agent	82.6	80.9	43.7	79.1	77.3	40.5
D-SECURE	83.5	81.9	45.1	79.8	78.2	43.4
ProgV-Trace	**87.1**	**85.8**	**66.9**	**84.2**	**82.6**	**63.1**

5.1 RQ1: Main Comparison on GroundLie360

Table 1 presents the end-to-end comparison on GroundLie360. ProgV-Trace achieves the highest performance across all six metrics, with Resp-NDCG reaching 71.4%—surpassing D-SECURE by 24.2 points, confirming that program-guided verification produces substantially better attribution than post-hoc explanations. The Prog-SR of 92.3% indicates well-formed program generation. Among baselines, D-SECURE achieves the second-best classification via dual-source reasoning, while REFORM leads in grounding metrics.

5.2 RQ2: Cross-Dataset Generalization

Table 2 evaluates generalization. ProgV-Trace consistently outperforms all baselines, with Resp-NDCG advantages of 21.8 and 19.7 points over D-SECURE on FakeVV and FakeSV respectively, confirming that the program-guided attribution gap persists across benchmarks.

5.3 RQ3: Program-Hard Challenge Protocol

Table 3 evaluates robustness on three challenge splits. All baselines suffer 8–20 point F1 drops compared to standard benchmarks [73], while ProgV-Trace maintains the smallest gap—most pronounced on Program-hard (+12.6 F1 over D-SECURE), where multi-step verification is required. The Resp-NDCG advantage widens further on challenge splits, confirming that program-guided reasoning is robust when surface-level shortcuts are removed.

5.4 RQ4: Ablation Study

Table 4 presents ablation results. Removing the verification program causes the largest overall degradation (−4.1 F1, −19.1 NDCG), confirming that program-guided verification is the most critical component. Direct chain-of-thought reasoning recovers some F1 but still loses 16.6 NDCG points, so the structured DSL representation provides attribution benefits beyond free-text reasoning. Removing graph expansion or replacing it with flat retrieval further

Table 3: Program-Hard Challenge Protocol. Fresh: >3 months after cutoff. Coordinated: 2 annotators (κ=0.84). Prog-hard: ≥2 verification steps. 150–200 samples/split; p<0.05 (bootstrap, 1000 iterations).

Method	Fresh	Coord.	Prog-hard
F1 Score (%)			
DEFAME [6]	71.2	65.8	60.4
CMIE [25]	68.9	67.4	58.1
T2Agent [9]	72.4	71.2	64.8
D-SECURE [4]	73.6	74.1	66.3
ProgV-Trace	**81.2**	**82.5**	**78.9**
Resp-NDCG (%)			
DEFAME†	33.7	30.1	28.9
CMIE†	32.1	31.4	27.5
T2Agent†	35.8	36.1	33.2
D-SECURE†	37.2	38.6	35.7
ProgV-Trace	**60.8**	**64.3**	**58.1**

Table 4: Ablation study on GroundLie360. Each variant removes one key component.

Variant	F1	T-F1	T-IoU	NDCG	P-SR
Full ProgV-Trace	**87.2**	**72.8**	**63.1**	**71.4**	**92.3**
w/o Verif. Program	83.1	66.4	56.8	52.3	—
Direct CoT only	84.6	67.9	58.2	54.8	—
w/o Graph Expansion	84.2	64.1	55.3	56.7	89.1
Flat Retrieval	83.8	62.7	54.5	53.4	88.5
w/o Intervention	85.6	70.9	61.8	46.8	91.5
Attn. Score only	86.1	71.5	62.3	49.6	91.8
Student (distilled)	85.3	71.1	61.7	68.7	91.2
LLaMA-3-70B backbone	85.4	70.6	60.8	65.8	84.1
Video-only	72.4	—	50.6	39.8	85.2
Text-only	80.8	65.3	—	53.1	90.7

degrades Token-F1 and Temporal-IoU, confirming that program-guided provenance expansion improves both grounding and evidence organization.

Removing counterfactual intervention drops NDCG to 46.8%, and the "Attn. Score only" variant reaches only 49.6% against 71.4% for interventional scoring: attention spreads weight across co-occurring tokens, whereas intervention measures the verdict shift from a targeted replacement, producing sharper attribution.

The distilled student retains 97.8% of teacher F1 and 96.2% of Resp-NDCG, and a LLaMA-3-70B [54] backbone degrades gracefully (−1.8 F1, −5.6 NDCG, Prog-SR 84.1%), confirming generalizability beyond proprietary models. Video-only drops F1 to 72.4% and NDCG by 31.6 points, so both modalities are essential. The "Attn. Score only" ablation uses the same post-hoc protocol [46] as all baselines, isolating the attribution method effect; on correctly classified samples, responsibility scores agree with human annotations 84.9% on average (78.2%–91.1% across five manipulation types).

5.5 RQ5: Qualitative Analysis

Figure 3 illustrates ProgV-Trace on a recontextualized protest video captioned "Lagos, March 2024". The 7-step program retrieves geolocated evidence placing the event in São Paulo, identifies temporal and spatial conflicts, and applies `INTERVENE` to flip the verdict probability from 0.96 to 0.28, assigning R=0.83 to the location fabrication and R=0.54 to the temporal mismatch—whereas D-SECURE detects the post as false but cannot localize the fabrication.

Table 5: Efficiency comparison. "Teacher": full LLM pipeline. "Student": distilled verifier.

Method	ms/sample	Mem (GB)	Params (M)
FactGuard	**142**	**4.8**	**312**
T2Agent	318	7.4	—
D-SECURE	186	5.6	347
DEFAME	238	6.2	425
ProgV-Trace (Teacher)	485	8.1	—
ProgV-Trace (Student)	196	5.9	356

5.6 RQ6: Efficiency and Deployment

Table 5 compares computational cost. The teacher pipeline (485 ms/sample) pays for multi-step program execution, while the distilled student runs at 196 ms on 5.9 GB—comparable to D-SECURE (186 ms)—at the accuracy retention reported above.

5.7 RQ7: Error Analysis

ProgV-Trace's failures on GroundLie360 fall into four types: coordinated forgeries (29%), hardest because deliberately consistent cross-modal content leaves no internal inconsistency to detect; temporal mismatch (27%) and location ambiguity (24%), from contextually plausible visuals and ambiguous geographic cues; and cross-modal cover errors (20%), where authentic content in one modality masks fabrication in another. The CoT fallback recovers 68% of program failures.

6 Conclusion

We introduced L-EMMA, a task extending multimodal misinformation detection to quantified interventional responsibility attribution, and ProgV-Trace, a framework whose LLM-authored verification programs orchestrate evidence retrieval, provenance graph expansion, and counterfactual interventions. On GroundLie360, FakeVV, and FakeSV it improves detection, grounding, and attribution together, and the Program-Hard Challenge Protocol confirms robust generalization where existing methods degrade substantially.

Beyond accuracy, ProgV-Trace makes the verification process explicit and inspectable, letting moderators trace which evidence was retrieved and how responsibility was attributed, and the distilled student verifier deploys at 196 ms on a single 8 GB GPU while retaining over 96% of the teacher's attribution quality. Effectiveness depends on both LLM program quality and evidence retrieval coverage [17]; extending the DSL to multi-turn conversational misinformation and non-English content remains an important direction. Code and data will be released.

Acknowledgments

This work was supported by the Science and Technology Project of China Southern Power Grid Co., Ltd. (ZBKJXM20220010). We thank the anonymous reviewers for their constructive feedback, which helped improve this paper. Generative AI tools were used to assist with editing the prose and with drafting and debugging

experiment code; the authors take full responsibility for the content of this article.

References

[1] Sara Abdali et al. 2024. Multi-Modal Misinformation Detection: Approaches, Challenges, and Opportunities. In *Proceedings of the 47th International ACM SIGIR Conference*.

[2] Sahar Abdelnabi, Rakibul Hasan, and Mario Fritz. 2022. Open-Domain, Content-based, Multi-modal Fact-checking of Out-of-Context Images via Online Resources. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

[3] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, et al. 2022. Flamingo: a Visual Language Model for Few-Shot Learning. arXiv:2204.14198.

[4] Samudi Amarasinghe, Gagandeep Singh, and Priyanka Singh. 2026. D-SECURE: Dual-Source Evidence Combination for Unified Reasoning in Misinformation Detection. arXiv:2602.14441.

[5] Shivangi Aneja, Chris Bregler, and Matthias Nießner. 2023. COSMOS: Catching Out-of-Context Misinformation with Self-Supervised Learning. In *Proceedings of the IEEE/CVF CVPR*.

[6] Dennis Braun et al. 2025. DEFAME: Dynamic Evidence-based FAct-checking with Multimodal Experts. arXiv:2412.10510.

[7] Mark Chen et al. 2021. Evaluating Large Language Models Trained on Code. arXiv:2107.03374.

[8] Yixuan Chen, Dongsheng Li, Peng Zhang, Jie Sui, Qin Lv, Lu Tun, and Li Shang. 2022. Cross-modal Ambiguity Learning for Multimodal Fake News Detection. In *Proceedings of the Web Conference*.

[9] Xing Cui, Yueying Zou, Zekun Li, Peipei Li, Xinyuan Xu, Xuannan Liu, and Huaibo Huang. 2026. T^2Agent: A Tool-augmented Multimodal Misinformation Detection Agent with Monte Carlo Tree Search. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

[10] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of NAACL-HLT*.

[11] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. 2021. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations*.

[12] Sharad Duwal, Mir Nafis Sharear Shopnil, Abhishek Tyagi, and Adiba Mahbub Proma. 2025. Evidence-Grounded Multimodal Misinformation Detection with Attention-Based GNNs. arXiv:2505.18221.

[13] Zhijiang Guo, Michael Schlichtkrull, and Andreas Vlachos. 2022. A Survey on Automated Fact-Checking. In *Transactions of the Association for Computational Linguistics*, Vol. 10. 178–206.

[14] Tanmay Gupta and Aniruddha Kembhavi. 2023. Visual Programming: Compositional Visual Reasoning without Training. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

[15] Will Hamilton, Zhitao Ying, and Jure Leskovec. 2017. Inductive Representation Learning on Large Graphs. In *Advances in Neural Information Processing Systems*.

[16] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the Knowledge in a Neural Network. arXiv:1503.02531.

[17] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, et al. 2024. A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions. arXiv:2311.05232.

[18] Kalervo Järvelin and Jaana Kekäläinen. 2002. Cumulated Gain-Based Evaluation of IR Techniques. *ACM Transactions on Information Systems* 20, 4 (2002), 422–446.

[19] Zhiwei Jin, Juan Cao, Han Guo, Yongdong Zhang, and Jiebo Luo. 2017. Multimodal Fusion with Recurrent Neural Networks for Rumor Detection on Microblogs. In *Proceedings of the 25th ACM International Conference on Multimedia*.

[20] Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Judy Hoffman, Li Fei-Fei, C. Lawrence Zitnick, and Ross Girshick. 2017. Inferring and Executing Programs for Visual Reasoning. In *Proceedings of the IEEE/CVF ICCV*.

[21] Dhruv Khattar, Jaipal Singh Goud, Manish Gupta, and Vasudeva Varma. 2019. MVAE: Multimodal Variational Autoencoder for Fake News Detection. In *The World Wide Web Conference*.

[22] Thomas N. Kipf and Max Welling. 2017. Semi-Supervised Classification with Graph Convolutional Networks. In *International Conference on Learning Representations*.

[23] Jie Lei, Licheng Yu, Tamara L. Berg, and Mohit Bansal. 2020. TVR: A Large-Scale Dataset for Video-Subtitle Moment Retrieval. In *Proceedings of ECCV*.

[24] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems*.

[25] Fanxiao Li, Jiaying Wu, Canyuan He, and Wei Zhou. 2025. CMIE: Combining MLLM Insights with External Evidence for Explainable Out-of-Context Misinformation Detection. arXiv:2505.23449.

[26] Haiyang Li, Yaxiong Wang, Shengeng Tang, Lianwei Wu, Lechao Cheng, and Zhun Zhong. 2025. Towards Unified Multimodal Misinformation Detection in Social Media: A Benchmark Dataset and Baseline. arXiv:2509.25991.

[27] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023. BLIP-2: Bootstrapping Language-Image Pre-Training with Frozen Image Encoders and Large Language Models. In *International Conference on Machine Learning*.

[28] Yiheng Li, Yang Yang, Zichang Tan, Huan Liu, Weihua Chen, Xu Zhou, and Zhen Lei. 2025. Unleashing the Potential of Consistency Learning for Detecting and Grounding Multi-Modal Media Manipulation. arXiv:2506.05890.

[29] Zehao Li, Hongwei Yu, Hao Jiang, Qiang Sheng, Yilong Xu, Baolong Bi, Yang Li, Zhenlong Yuan, et al. 2026. FactGuard: Agentic Video Misinformation Detection via Reinforcement Learning. arXiv:2602.22963.

[30] Moyang Liu, Kaiying Yan, Yukun Liu, Ruibo Fu, Zhengqi Wen, Xuefei Liu, and Chenxing Li. 2025. Deconfounded Reasoning for Multimodal Fake News Detection via Causal Intervention. arXiv:2504.09163.

[31] Xuannan Liu et al. 2024. MMFakeBench: A Mixed-Source Multimodal Misinformation Detection Benchmark. arXiv:2406.08772.

[32] Ilya Loshchilov and Frank Hutter. 2019. Decoupled Weight Decay Regularization. arXiv:1711.05101.

[33] Scott M. Lundberg and Su-In Lee. 2017. A Unified Approach to Interpreting Model Predictions. In *Advances in Neural Information Processing Systems*.

[34] Grace Luo, Trevor Darrell, and Anna Rohrbach. 2021. NewsCLIPpings: Automatic Generation of Out-of-Context Multimodal Media. In *Proceedings of EMNLP*.

[35] Jiayuan Mao, Chuang Gan, Pushmeet Kohli, Joshua B. Tenenbaum, and Jiajun Wu. 2019. The Neuro-Symbolic Concept Learner: Interpreting Scenes, Words, and Sentences From Natural Supervision. In *International Conference on Learning Representations*.

[36] Kai Nakamura, Sharon Levy, and William Yang Wang. 2020. r/Fakeddit: A New Multimodal Benchmark Dataset for Fine-grained Fake News Detection. In *Proceedings of LREC*.

[37] Yulei Niu, Kaihua Tang, Hanwang Zhang, Zhiwu Lu, Xian-Sheng Hua, and Ji-Rong Wen. 2021. Counterfactual VQA: A Cause-Effect Look at Language Bias. In *Proceedings of the IEEE/CVF CVPR*.

[38] Haoran Ou, Gelei Deng, Xingshuo Han, Jie Zhang, Han Qiu, Shangwei Guo, and Tianwei Zhang. 2025. Beyond Retrieval: Improving Evidence Quality for LLM-based Multimodal Fact-Checking. arXiv:2505.03135.

[39] Stefanos-Iordanis Papadopoulos, Christos Koutlis, Symeon Papadopoulos, and Panagiotis C. Petrantonakis. 2023. VERITE: A Robust Benchmark for Multimodal Misinformation Detection Accounting for Unimodal Bias. arXiv:2304.14133.

[40] Stefanos-Iordanis Papadopoulos, Christos Koutlis, Symeon Papadopoulos, and Panagiotis C. Petrantonakis. 2024. Similarity over Factuality: Are We Making Progress on Multimodal Out-of-Context Misinformation Detection? arXiv:2407.13488.

[41] Judea Pearl. 2009. *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press.

[42] Peng Qi, Juan Cao, Tianyun Yang, Junbo Guo, and Jintao Li. 2021. Multimodal Fusion with Co-Attention Networks for FakeSV Detection. In *Findings of ACL*.

[43] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning Transferable Visual Models from Natural Language Supervision. In *International Conference on Machine Learning*.

[44] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. 2016. "Why Should I Trust You?": Explaining the Predictions of Any Classifier. arXiv:1602.04938.

[45] Michael Schlichtkrull, Thomas N. Kipf, Peter Bloem, Rianne van den Berg, Ivan Titov, and Max Welling. 2018. Modeling Relational Data with Graph Convolutional Networks. In *European Semantic Web Conference*.

[46] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. 2017. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. In *Proceedings of the IEEE International Conference on Computer Vision*.

[47] Rui Shao, Tianxing Wu, and Ziwei Liu. 2023. Detecting and Grounding Multi-Modal Media Manipulation. In *Proceedings of the IEEE/CVF CVPR*. Introduces the DGM4 dataset and task.

[48] Jinjie Shen, Yaxiong Wang, Lechao Cheng, Nan Pu, and Zhun Zhong. 2025. Beyond Artificial Misalignment: Detecting and Grounding Semantic-Coordinated Multimodal Manipulations. arXiv:2509.12653.

[49] Kai Shu, Limeng Cui, Suhang Wang, Dongwon Lee, and Huan Liu. 2019. dEFEND: Explainable Fake News Detection. In *Proceedings of the 25th ACM SIGKDD*.

[50] Ishika Singh, Valts Blukis, Arsalan Mousavian, Ankit Goyal, Danfei Xu, Jonathan Tremblay, Dieter Fox, Jesse Thomason, and Animesh Garg. 2023. ProgPrompt: Generating Situated Robot Task Plans using Large Language Models. arXiv:2209.11302.

[51] Dídac Surís, Sachit Menon, and Carl Vondrick. 2023. ViperGPT: Visual Inference via Python Execution for Reasoning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.

[52] Kaihua Tang, Yulei Niu, Jianqiang Huang, Jiaxin Shi, and Hanwang Zhang. 2020. Unbiased Scene Graph Generation from Biased Training. In *Proceedings of the IEEE/CVF CVPR*.

[53] James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. 2018. FEVER: a Large-scale Dataset for Fact Extraction and VERification. In *Proceedings of NAACL-HLT*.

[54] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. 2023. LLaMA: Open and Efficient Foundation Language Models. arXiv:2302.13971.

[55] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is All You Need. In *Advances in Neural Information Processing Systems*.

[56] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. 2018. Graph Attention Networks. In *International Conference on Learning Representations*.

[57] Junxi Wang, Yaxiong Wang, Lechao Cheng, and Zhun Zhong. 2025. FakeSV-VLM: Taming VLM for Detecting Fake Short-Video News via Progressive Mixture-Of-Experts Adapter. arXiv:2508.19639.

[58] Tan Wang, Jianqiang Huang, Hanwang Zhang, and Qianru Sun. 2020. Visual Commonsense R-CNN. In *Proceedings of the IEEE/CVF CVPR*.

[59] Yaqing Wang, Fenglong Ma, Zhiwei Jin, Ye Yuan, Guangxu Xun, Kishlay Jha, Lu Su, and Jing Gao. 2018. EANN: Event Adversarial Neural Networks for Multi-Modal Fake News Detection. In *Proceedings of the 24th ACM SIGKDD*.

[60] Yaqing Wang, Fenglong Ma, Haoyu Wang, Kishlay Jha, and Jing Gao. 2021. Multimodal Emergent Fake News Detection via Meta Neural Process Networks. In *Proceedings of the 27th ACM SIGKDD*.

[61] Claire Wardle and Hossein Derakhshan. 2017. Information Disorder: Toward an Interdisciplinary Framework for Research and Policymaking. Council of Europe Report DGI(2017)09.

[62] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. arXiv:2201.11903.

[63] Keyulu Xu, Weihua Hu, Jure Leskovec, and Stefanie Jegelka. 2019. How Powerful are Graph Neural Networks?. In *International Conference on Learning Representations*.

[64] Bingjian Yang, Danni Xu, Kaipeng Niu, Wenxuan Liu, Zheng Wang, and Mohan Kankanhalli. 2025. A New Dataset and Benchmark for Grounding Multimodal Misinformation. arXiv:2509.08008. Introduces the GroundLie360 dataset and the FakeMark baseline.

[65] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. 2023. Tree of Thoughts: Deliberate Problem Solving with Large Language Models. arXiv:2305.10601.

[66] Zhenrui Yue, Huimin Zeng, Ziyi Kou, Lanyu Shang, and Dong Wang. 2022. Contrastive Domain Adaptation for Early Misinformation Detection: A Case Study on COVID-19. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*.

[67] Fanrui Zhang, Dian Li, Qiang Zhang, Jun Chen, Gang Liu, Junxiong Lin, Jiahong Yan, Jiawei Liu, et al. 2025. Fact-R1: Towards Explainable Video Misinformation Detection with Deep Reasoning. arXiv:2505.16836.

[68] Fanrui Zhang, Dian Li, Qiang Zhang, Jun Chen, Gang Liu, Junxiong Lin, Jiahong Yan, Jiawei Liu, et al. 2025. Fact-R1: Towards Explainable Video Misinformation Detection with Deep Reasoning. arXiv:2505.16836. Introduces the FakeVV dataset.

[69] Yuchen Zhang, Yaxiong Wang, Kecheng Han, Yujiao Wu, Lianwei Wu, Li Zhu, and Zhedong Zheng. 2026. Process Over Outcome: Cultivating Forensic Reasoning for Generalizable Multimodal Manipulation Detection. arXiv:2603.01993.

[70] Zhenxing Zhang, Yaxiong Wang, Lechao Cheng, Zhun Zhong, Dan Guo, and Meng Wang. 2024. ASAP: Advancing Semantic Alignment Promotes Multi-Modal Manipulation Detecting and Grounding. arXiv:2412.12718.

[71] Xinyi Zhou, Arjun Mukherjee, Reza Shivakumar, and Dan Goldwasser. 2023. Multimodal Fake News Detection: A Survey of Deep Learning Techniques. In *ACM Computing Surveys*, Vol. 55.

[72] Xinyi Zhou, Jindi Wu, and Reza Zafarani. 2020. SAFE: Similarity-Aware Multi-modal Fake News Detection. In *Proceedings of PAKDD*.

[73] Yongchun Zhu, Qiang Sheng, Juan Cao, Shuokai Li, Danding Wang, and Fuzhen Zhuang. 2022. Generalizing to the Future: Mitigating Entity Bias in Fake News Detection. In *Proceedings of the 45th International ACM SIGIR Conference*.

[74] Linlin Zong, Wenmin Lin, Jiahui Zhou, Xinyue Liu, Xianchao Zhang, Bo Xu, and Shimin Wu. 2025. Text-Guided Fine-grained Counterfactual Inference for Short Video Fake News Detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39.